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## **CE-6002-CBGS**

### **B.E. VI Semester**

Examination, December 2020

## **Choice Based Grading System (CBGS)**

### **Structural Design - I (RCC)**

*Time : Three Hours*

*Maximum Marks : 70*

- Note:** i) Attempt any five questions.  
ii) All questions carry equal marks.  
iii) Assume suitability data.

1. a) Write down the Assumptions in the working stress method. Differentiate between working stress method and limit state method. 7  
b) Find the depth of Neutral axis and percentage of steel for a balanced section of a singly reinforced beam using  
i) M-20 and plain mild steel bars.  
ii) M-20 and Fe-415 bars. 7
2. a) Determine the moment of resistance of beam having the following data.  $b = 350 \text{ mm}$ ,  $d = 900 \text{ mm}$ ,  $d' = 50 \text{ mm}$ , tension reinforcement = 5-20 mm dia bars, compression reinforcement = 2-20 mm dia bars. 7  
b) An R.C.C beam  $250 \text{ mm} \times 400 \text{ mm}$  effective is carrying a uniformly distributed load of  $16 \text{ kN/m}$ . The beam is reinforced with 4 bars of 22 mm dia. The clear span of the beam is 4m. Design the shear reinforcement. Use M-20 and Fe-250. 7

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3. Design a simply supported roof slab for a room  $7.5 \text{ m} \times 3.5 \text{ m}$  clear in size. The slab is carrying an imposed load of  $5 \text{ kN/m}^2$ . Use M-20 mix and Fe415 steel. 14
4. Design a square footing of uniform thickness for an axially loaded column of  $450 \text{ mm} \times 450 \text{ mm}$  size. The safe bearing capacity of soil is  $190 \text{ kN/m}^2$ . Load on column is  $850 \text{ kN}$ -M20 Mix and Fe-415 steel 14
5. a) Explain the structural behaviour of a stair case. 7  
b) Explain the IS code recommendations for the effective span of the stair slab when landing spans perpendicular to the flight. 7
6. A reinforced concrete beam is simply supported over a clear span of  $6 \text{ m}$ . The beam carries a superimposed load of  $10 \text{ kN/m}$ . Design the beam if the width of the beam is  $300 \text{ mm}$ . Use M20 mix and Fe 415 steel. The beam is resting on  $400 \text{ mm}$  thick wall. 14
7. a) Determine the ultimate moment of resistance of a  $150 \text{ mm}$  thick slab reinforced with  $10 \text{ mm } \phi$  bars at  $200 \text{ C/C}$ . The effective cover is  $25 \text{ mm}$ . Use M20 concretes and Fe 415 steel. 7  
b) What are Footings? How are they classified? 7
8. Design a cantilever beam projecting  $2.5 \text{ m}$  beyond the fixed end and carrying a UDL of  $10 \text{ kN/m}$ . Use M20 mix and Fe 415 steel. 14

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